

Lesson 3 Outline

- System Equations
- Differential and Difference Equations
- Partial and Ordinary Differential Equation
- Time Variability and Time Invariance
- Linear and Non-Linear Differential and Difference Equations

SYSTEM EQUATIONS

A property common to all basic laws of physics is that certain fundamental quantities can be defined by numerical values. The physical laws define relationships between these fundamental quantities and are usually represented by equations.

EXAMPLE: The scalar version of Newton's second law states that, if a force of magnitude f is applied to a mass of M units, the acceleration a of the mass is related to f by the equation f = Ma.

EXAMPLE: Ohm's law states that, if a voltage of magnitude V is applied across a resistor of R units, the current i through the resistor is related to V by the equation V = Ri.

Many nonphysical laws can also be represented by equations. EXAMPLE: The compound interest law states that, if an amount P(0) is deposited for n equal periods of time at an interest rate i for each time period, the amount will grow to a value of P(n) = P(0)(1 + i)n .

Differential Equations and Difference Equations

Two classes of equations with broad application in the description of systems are differential equations and difference equations.

Definition 3.1: A differential equation is any algebraic or transcendental equality which involves either differentials or derivatives. Operations like addition or subtraction have a different special representation that looks like a small circle referred to as a summing point with the appropriate plus or minus sign associated with the arrows entering the circle.

Differential equations are useful for relating rates of change of variables and other parameters.

EXAMPLE: Newton's second law can be rewritten alternatively as a relationship between force f, mass M, and the rate of change of the velocity v of the mass with respect to time t, that is, f = M(dv/dt).

EXAMPLE: Ohm's law can be rewritten alternatively as a relationship between voltage resistance R, and the time rate of passage of charge through the resistor, that is, V = R(dq/dt).

Definition 3.2: A difference equation is an algebraic or transcendental equality which involves more than one value of the dependent variable(s) corresponding to more than one value of at least one of the independent variable(s). The dependent variables do not involve either differentials or derivatives.

Difference equations are useful for relating the evolution of variables (or parameters) from one discrete instant of time (or other independent variable) to another.

EXAMPLE: The compound interest law can be written alternatively as a difference equation relationship between P(k), the amount of money after k periods of time, and P(k+1), the amount of money after k+1 periods of time, that is, P(k+1) = (1+i)P(k).

Partial and Ordinary Differential Equations Definition 3.3: A partial differential equation is an equality involving one or more dependent and two or more independent variables, together with partial derivatives of the dependent with respect to the independent variables.

Definition 3.4: An ordinary (total) differential equation is an equality involving one or more dependent variables, one independent variable, and one or more derivatives of the dependent variables with respect to the independent variable.

EXAMPLE: The diffusion equation ∂T/∂x = k(∂T/∂t) is a partial differential equation. T = T(x, t) is the dependent variable, which represents the concentration of some quantity at some position and some time in the object. The independent variable x defines the position in the object, and the independent variable t defines the time.

EXAMPLE: Ohm's law is an ordinary differential equation v = R(dq/dt). The charge q = q(t) and the voltage v = v(t) are dependent variables, and the time t is the independent variable. EXAMPLE: Newton's second law is an ordinary differential equation f = M(dv/dt). The velocity v = v(t) and the force f = f(t) are dependent

variables, and the time t is the independent variable.

Note that the time is the only independent variable, unless otherwise specified. This variable is normally designated t, except that in difference equations the discrete variable k is often used, as an abbreviation for the time instant tk , that is, y(k) is used instead of y(tk), etc. A term of a differential or difference equation consists of products and/or quotients of explicit functions of the independent variable, the dependent variables, and, for differential equations, derivatives of the dependent variables.

Time variability and Time invariance

Definition 3.5: A time-variable equation is an equation in which one or more terms depend explicitly on the independent variable time.

Definition 3.6: A time-invariant equation is an equation in which none of the terms depends explicitly on the independent variable time.

EXAMPLE: The difference equation ky(k+2) + y(k) = u(k), where u and y are dependent variables, is time-variable because the term ky(k+2) depends explicitly on the coefficient k, which represents the time tk. EXAMPLE: Any differential equation similar to the form below, (where the coefficients a0, a1, a2,..., an, b0, b1, b2,..., bm, are constants) is time-invariant. The equation depends implicitly on t, via the dependent variables u and y and their derivatives.

Linear and Non-linear Differential and Difference Equations

Definition 3.7: A linear term is one which is first degree in the dependent variables and their derivatives.

Definition 3.8: A linear equation is an equation consisting of a sum of linear terms. All others are nonlinear equations.

If any term of a differential equation contains higher powers, products, or transcendental functions of the dependent variables, it is nonlinear. Such terms include (dy/dt)3 , u(dy/dt), and sin u, respectively.

For example, (5/cos t)(d2y/dt2) is a term of first degree in the dependent variable y, and 2uy3(dy/dt) is a term of fifth degree in the dependent variables u and y. EXAMPLE: The ordinary differential equations,

(dy/dt)2 + y = 0 and d2y/dt2 + cos y = 0

are nonlinear because (dy/dt)2 is second degree in the first equation, and cos y in the second equation is not first degree, which is true of all transcendental functions.

EXAMPLE: The difference equation,

y(k+2) + u(k+1) y(k+1) + y(k) = u(k)

in which u and y are dependent variables, is a nonlinear difference equation because u(k+1) y(k+1) is second degree in type of nonlinear equation is sometimes called bilinear in u and y.

EXAMPLE: Any difference equation in which the coefficients ai(k) and bi(k) depend only upon the independent variable k, is a linear difference equation.

EXAMPLE: Any ordinary differential equation where the coefficients ai(t) and bi(t) depend only upon the independent variable t, is a linear differential equation.

Differential Operator D and the Characteristic Equation

Consider the nth-order linear constant-coefficient differential equation.

$$\frac{d^n y}{dt^n} + a_{n-1}\frac{d^{n-1}y}{dt^{n-1}} + \cdots + a_1\frac{dy}{dt} + a_0 y = u$$

It is convenient to define a differential operator D

$$D \equiv \frac{d}{dt}$$

and more generally an nth order differential operator

Differential Operator

The differential equation,

$$\frac{d^n y}{dt^n} + a_{n-1}\frac{d^{n-1}y}{dt^{n-1}} +$$

can now be written down to this form,

$$D^n y + a_{n-1}D^{n-1}y +$$
$$(D^n + a_{n-1}D^{n-1} + \cdot$$
$$\cdot + a_1\frac{dy}{dt} + a_0 y = u$$

$$\cdot + a_1 Dy + a_0 y = u$$
$$+ a_1 D + a_0)y = u$$

Definition 3.9: The polynomial in D is called the characteristic polynomial.

$$(D^n + a_{n-1}D^{n-1} + \cdots + a_1 D + a_0)$$

Definition 3.10: The equation in D is called the characteristic equation.

The fundamental theorem of algebra states that the characteristic equation has exactly n solutions D = D1, D = D2,..., D = Dn. These n solutions (also called roots) are not necessarily distinct.

EXAMPLE: Consider the differential equation

$$\frac{d^2 y}{dt^2} + 3\frac{dy}{dt} + 2y = u$$

The characteristic polynomial is D2 + 3D + 2. The characteristic equation is D2 + 3D + 2 = 0, which has the two distinct roots, D = -1 and D = -2.

Linear Independence and Fundamental Sets

Definition 3.11: A set of n functions of time f1(t), f2(t),..., fn(t) is called linearly independent if the only set of constants c1, c2, c3, . . ., cn for which

$$c_1 f_1(t) + c_2 f_2(t) + \cdots + c_n f_n(t) = 0$$

for all t are the constants c1 = c2 = … = cn = 0. EXAMPLE: The functions t and t2 are linearly independent functions since

$$c_1 t + c_2 t^2 = (c_1 + c_2 t)t = 0$$

implies that c1/c2 = -t. There are no constants that satisfy this relationship.

A homogeneous nth order linear differential equation of the form:

$$\sum^{\cdot\cdot} a_i \frac{d^i y}{dt^i} = 0$$

has at least one set of n linearly independent solutions. Definition 3.12: Any set of n linearly independent solutions of a homogeneous nth-order linear differential equation is called a fundamental set.

There is no unique fundamental set. From a given fundamental set other fundamental sets can be generated by the following technique. Suppose that y1(t), y2(t),..., yn(t) is a fundamental set for an nth-order linear differential equation. Then a set of n functions z1(t), z2(t) ,..., zn(t) can be formed:

$$z_1(t) = \sum_{i-1} a1_i y_i(t), z_2(t) = \sum_{i-1} a2_i y_i(t), \ldots, z_n(t) = \sum_{i-1} an_i y_i(t)$$

Where ai are a set of n2 constants.

Each zi(t) is a solution of the differential equation. This set of n functions is a fundamental set if the determinant

EXAMPLE: The equation for simple harmonic motion,

has as a fundamental set

d2y/dt2 + w2y = 0

y1 = sin⁡t and y2 = cos⁡t.

A second fundamental set is,

$$z_1 = \cos\omega t + j\sin\omega t = e^{j\omega t}$$
$$z_2 = \cos\omega t - j\sin\omega t = e^{-j\omega t}$$



Distinct Roots: If the characteristic equation n

$$\sum a_i D^i = 0$$

has distinct roots D1, D2 ,..., Dn, then a fundamental set for the homogeneous equation

is the set of functions y1= eD1t , y2= eD2t ,..., yn = eDnt .

EXAMPLE: The differential equation

$$\frac{d^2y}{dt^2} + 3\frac{dy}{dt} + 2y = 0$$

has the characteristic equation D2 + 3D + 2 = 0 whose roots are D = D1 = -1 and D2 = -2.

A fundamental set for this equation is y1 = e-t and y2 = e-2t .

Repeated Roots: If the characteristic equation has repeated roots, then for each root Di of multiplicity ni (i.e., ni roots equal to Di) there are ni elements of the fundamental set eDit , teDit ,..., tni-1eDit . EXAMPLE: The equation

$$\frac{d^2y}{dt^2} + 2\frac{dy}{dt} + y = 0$$

with characteristic equation D2 + 2D + 1 = 0, has the repeated root D = -1 and a fundamental set consisting of e-t and te-t .

Solution of Linear Constant-Coefficient Ordinary Differential Equations

Consider the class of differential equations of the form:

$$\sum^{\cdot\cdot} a_i \frac{d^i y}{dt^i} = \sum^{\cdot\cdot\cdot} b_i \frac{d^i u}{dt^i}$$

where the coefficients ai and bi are constant, u = u(t) (the input) is a known time function, and y = y(t) (output) is the unknown solution of the equation. If this equation describes a physical system, then generally m ≤ n, and n is called the order of the differential equation.

To completely specify the problem so that a unique solution y(t) can be obtained, two additional items must be specified: (1) the interval of time over which a solution is desired and (2) a set of n initial conditions for y(t) and its first n - 1 derivatives. The time interval for the class of problems considered is defined by 0 ≤ t < +∞. The set of initial conditions is

$$y(0), \frac{dy}{dt}\Big|_{t=0}, \frac{d^2y}{dt^2}\Big|_{t=0}, \cdots, \frac{d^{n-1}y}{dt^{n-1}}$$

an initial value problem.

A problem defined over this interval and with these initial conditions is called

The solution of a differential equation of this class can be divided into two parts, a free response and a forced response. The sum of these two responses constitutes the total response, or solution y(t), of the equation.

The free response of a differential equation is the solution of the differential equation when the input u(t) is identically zero. If the input u(t) is identically zero, then the differential equation has the form:

$$\sum^{\cdot\cdot} a_i \frac{d^i y}{dt^i} = 0$$

The solution y(t) of such an equation depends only on the n initial conditions.

EXAMPLE: The solution of the homogeneous first-order differential equation dy/dt + y = 0 with initial condition y(0) = c, is y(t) = ce-t . This can be verified by direct substitution. ce-t is the free response of any differential equation of the form dy/dt + y = u with the initial condition y(0) = c.

The free response of a differential equation can always be written as a linear combination of the elements of a fundamental set.

That is, if y1(t), y2(t) ,. .. , yn(t) is a fundamental set, then any free response yn(t) of the differential equation can be represented as

$$y_a(t) = \sum c_i y_i(t)$$

where the constants ci are determined in terms of the initial conditions.

From the set of n algebraic equations,

$$y(0) = \sum_{i=1}^{n} c_i y_i(0), \frac{dy}{dt}|_{t=0} = \sum_{i=1}^{n} c_i \frac{dy_i}{dt}|_{t=0}, \ldots, \frac{d^{n-1}y}{dt^{n-1}}$$
$$= \sum_{i=1}^{n} c_i \frac{d^{n-1}y_i}{dt^{n-1}}|_{t=0}$$

The linear independence of the yi(t) guarantees that a solution to these equations can be obtained for C1, C2,..., Cn.

EXAMPLE: The free response ya(t) of the differential equation

$$\frac{d^2y}{dt^2} + 3\frac{dy}{dt} + 2y = u$$

with initial conditions y(0) = 0, (dy/dt)|t=0 = y'(0) = 1 is determined by letting where c1 and c2 are unknown coefficients and e-t and e-2t are a fundamental set for this equation.

then

$$y_a(0) = y(0) = c_1 + c_2$$
$$\frac{dy_a}{dt}|_{t=0} = \frac{dy}{dt}|_{t=0} = 1 = -c_1 - 2c_2$$

c1 = 1 and c2 = - 1. The free response is hence given by ya(t)= e-t - e-2t

The forced response yb(t) of a differential equation is the solution of the differential equation when all the initial conditions are identically zero.

$$y(0), \frac{dy}{dt}|_{t=0}, \frac{d^2y}{dt^2}|_{t=0}, \ldots, \frac{d^{n-1}y}{dt^{n-1}}$$

The implication of this definition is that the forced response depends only on the input u(t). The forced response for a linear constant-coefficient ordinary differential equation can be written in terms of a convolution integral,

where w(t -) is the weighting function (or kernel) of the differential

equation. This form of the convolution integral assumes that the weighting function describes a causal system. This assumption is maintained below. The weighting function of a linear constant-coefficient ordinary differential equation can be written as, where c1,..., cn are constants and the set of functions y1(t), y2(t) ,. .. , yn(t) is a fundamental set. It should be noted that wt is the free response of the differential equation and therefore requires n initial conditions for complete specification. This conditions fix the value of the constants c1,..., cn. The initial conditions in which all weighting function of the linear differentiating equations must satisfy are

Example: The weighting function for the differential equation

$$\frac{d^2y}{dt^2} + 3\frac{dy}{dt} + 2y = u$$

is a linear combination of e -t and e -2t (a fundamental set of the equation). That is,

$$w(t) = c_1 e^{-t} + c_2 e$$

c1 and c2 are determined from the two algebraic equations,

$$w(0) = c_1 + c_2$$
$$\frac{dw}{dt}|_{t=0} = 1 = -c_1 - 2c_2$$

The solution is c1 = 1, c2 = -1, and the weighting function is w(t) = e -t - e -2t

Example: For the differential equation

$$\frac{d^2y}{dt^2} + 3\frac{dy}{dt} + 2y = u$$

if u(t) = 1, then the forced response yb(t) of the equation,

$$y_b(t) = \int_0^t w(t-\tau)\,u(t)d\tau$$

$$y_b(t) = \int_0^t \left[e^{-(t-\tau)} + e^{-2(t-\tau)}\right] d\tau$$

$$y_b(t) = e^{-t}\int_0^t e^{\tau}d\tau - e^{-2t}\int_0^t e^{2\tau}d\tau$$

$$y_b(t) = \frac{1}{2}[1 - 2e^{-t} + e^{-2t}]$$

The total response of a linear constant coefficient differential equation is

the sum of the free response and forced response. Example: The total response y(t) of the differential equation

$$\frac{d^2y}{dt^2} + 3\frac{dy}{dt} + 2y =$$

initial condition y(0) = 0

$$y(t) = y_a(t) + y_b(t)$$

$$y_a(t) = e^{-t} - e^{-2}\ ; y_b(t) = \frac{1}{2}[1 - 2e^{-t} + e^{-2t}]$$

$$y(t) = e^{-t} - e^{-2t} + \frac{1}{2}[1 - 2e^{-t} + e^{-2t}]$$

and dy/dt|t=0 = y'(0) = 1 is the sum of the free response ya(t) and the forced response yb(t).

Steady State and Transient response

The Steady State and Transient Responses

The steady state response and transient response are another pair of quantities whose sum is equal to the total response. These terms are often used for specifying control system performance. They are defined as follows: Definition 3.13: The steady state response is that part of the total response which does not approach zero as time approaches infinity.

Definition 3.14: The transient response is that part of the total response which approaches zero as time approaches infinity.

Steady State and Transient response

Example: The total response y(t) of the differential equation

$$\frac{d^2y}{dt^2} + 3\frac{dy}{dt} + 2y =$$

initial condition y(0) = 0 and dy/dt|t=0 = y'(0) = 1 is

$$y(t) = e^{-t} - e^{-2t} + \frac{1}{2}[1 - 2e^{-t} + e^{-2t}]$$

$$y(t) = \frac{1}{2} - \frac{1}{2}e^{-2t}$$

Therefore the steady state yss = ½ and the transient response is yt = ½ e-2t since,

$$\lim_{t\to\infty} \frac{1}{2}e^{-2t} = 0$$

Steps, Ramps, and Impulses

In the study of control systems and the equations which describe them, a particular family of functions called singularity functions is used extensively. Each member of this family is related to the others by one or more integrations or differentiations. The three most widely used singularity functions are the unit step, the unit impulse, and the unit ramp.

Definition 3.16: A unit ramp function is the integral of a unit step function

Δ → Δ

The limit pair may be abbreviated by t 0+ , meaning that t approaches zero from the right. The quotient in brackets represents a rectangle of height 1/t and width t. The limiting process produces a function whose height approaches infinity and width approaches zero.

The area under the curve is equal to 1 for all values of t . That is,

$$\int_{-\infty}^{\infty} \delta(t)dt =$$

The unit impulse function has a very important property called Screening

Property.

Screening Property:

The integral of the product of a unit impulse function (t – t0) and a function f(t), continuous at t = t0 over an interval which includes t0, is equal to the

function f(t) evaluated at t0, that is,

Definition 3.28: The unit impulse response of a system is the output y(t) of the system when the input u(t) = (t) and all initial conditions are zero.

EXAMPLE: If the input-output relationship of a linear system is given by the convolution integral

$$y(t) = \int_0^t w(t-\tau)u(\tau)\,d\tau$$

then the unit impulse response y(t) of the system is

$$y_\delta(t) = \int_{-\infty}^{\infty} w(t-\tau)\delta(\tau)\,d\tau = w(t)$$

since w(t - 𝜏) = 0 for 𝜏 > t, 𝜏(𝜏) = 0 for 𝜏 < 0, and the screening property of the unit impulse has been used to evaluate the integral.

Definition 3.29: The unit step response is the output y(t) when the input u(t) = 1(t) and all initial conditions are zero.

Definition 3.20: The unit ramp response is the output y(t) when the input u(t) = t for t > 0, u(t) = 0 for t 𝜏 0, and all initial conditions are zero.



Thank you!